



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

This report talks about a machine learning-based pipeline for segmenting the Rectus Femoris muscle in ultrasound images and estimating Anatomical Cross-Sectional Area (ACSA). A Random Forest classifier is trained using pixel-wise intensity and structural features. To improve the model, we reduce redundancy in the feature set by removing less useful features based on importance and correlation.

We performed a diagnostic experiment to understand how much the model depends on spatial features. For evaluation, we focused on Dice, IoU, and pixel area to avoid reliance on ground truth calibration. ACSA in cm^2 is reported only as a secondary estimate and is clearly treated as dataset-calibrated. Overall, we tried to perform experiments to check which of our features are redundant, to better set our feature matrix.

Contents

1	What We Have Done	4
2	Results	4
3	What We Aspire to Do	6

1 What We Have Done

- We looked deeply into our segmentation pipeline and identified that the model may be relying too much on spatial features rather than actual image information.
- To check if this is correct or not, we designed an experiment where spatial features (such as distance to center and boundary) are removed and the model is retrained.
- We analysed feature importance and noticed that some features were contributing very little while also being highly redundant.
- Based on this, we simplified the feature set by removing `gx_s1`, `gy_s1`, `gmag_s1`, and `std_7`, while keeping more useful features like `gmag_s3` and `laplacian`.
- We made sure that this feature pruning step is applied only after the spatial diagnostic, so that the importance ranking is not biased.
- We also examined how ACSA is currently computed and realised that it depends on ground truth statistics through dataset-level calibration.
- To address this, we shifted our focus towards evaluation metrics that are independent of ground truth, such as Dice, IoU, and pixel area.

2 Results

Table 1: Effect of Removing Spatial Features

Metric	Original (with spatial)	Diagnostic (no spatial)	Drop
Val Dice	0.7475	0.6330	+0.1145
Test Dice	0.6741	0.6025	+0.0716

Interpretation:

- Drop 0.05–0.17: Partial spatial dependency.
- Model uses both appearance and position.
- Pipeline is partially valid. Spatial features provide meaningful boost.

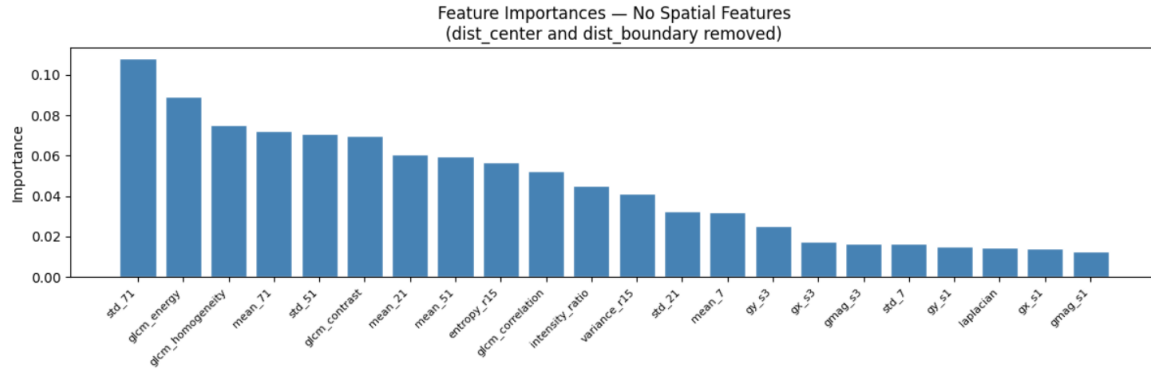


Figure 1: Feature importance



Figure 2: Worst Performance



Figure 3: Best Performance

3 What We Aspire to Do

- Based on the experiment results, we will finalise the pruned feature set and retrain the model accordingly.
- We will evaluate the updated pipeline using Dice, IoU, and pixel area as the main metrics to ensure the results are independent and reliable.
- ACSA (cm^2) will still be reported, but only as a secondary estimate and clearly marked as dataset-calibrated.
- We will also observe how feature pruning affects model stability and overall performance.
- Finally, we aim to refine the pipeline so that it is simpler, less biased, and more interpretable.

References

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